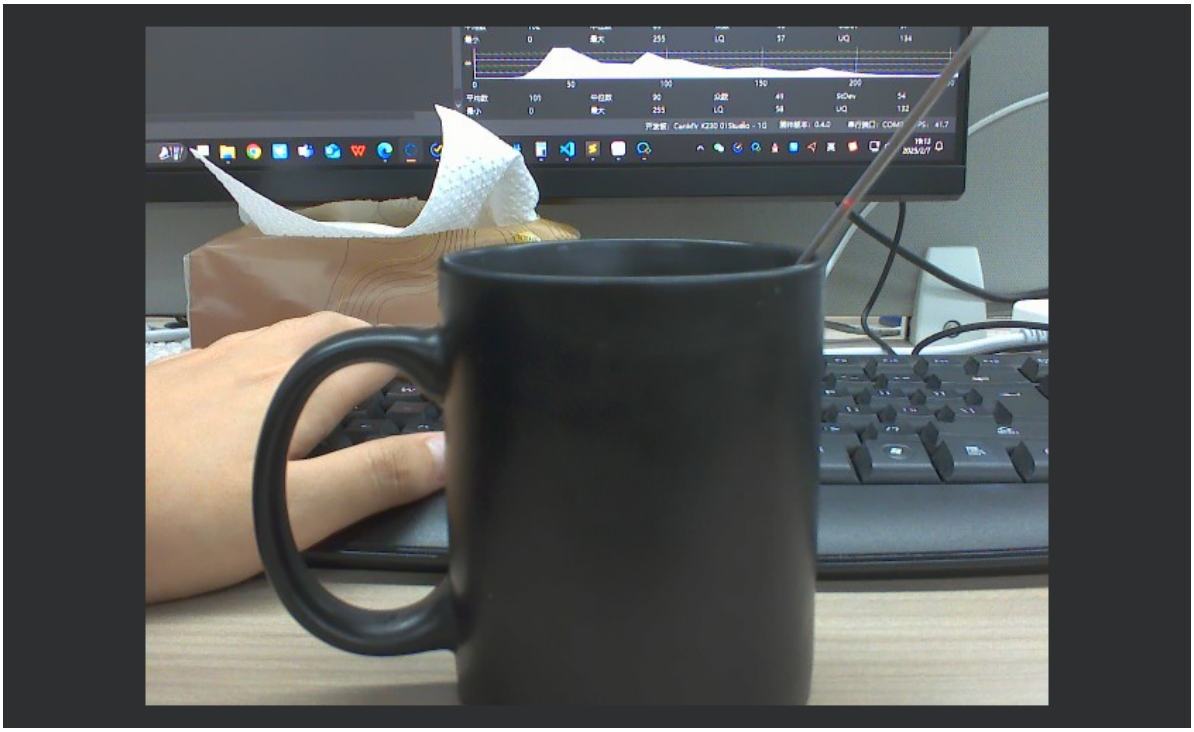
摄像头图像翻转

摄像头图像翻转
- 摄像头水平镜像（翻转）
  - API描述
  - 例程
- 摄像头垂直翻转
  - API 描述
  - 例程

未翻转前的摄像头捕获到的图像如图所示：



摄像头水平镜像（翻转）

API描述

sensor.set_hmirror

描述

配置图像传感器是否进行水平镜像。

语法

```
sensor.set_hmirror(enable)
```

参数

参数名称	描述	输入/输出
enable	`True` 开启水平镜像功能 `False` 关闭水平镜像功能	输入

返回值

返回值	描述
无	

举例

```
sensor.set_hmirror(True)
```

例程

代码位于 [源码汇总 / 02.Basic /18.1_sensor_hmirror.py]

```
"""
Camera preview demo
摄像头预览演示

This script initializes camera sensor, displays preview and handles cleanup
本脚本初始化摄像头传感器、显示预览并处理清理工作
"""

import sys
import uos as os
import time
from media.sensor import *
from media.display import *
from media.media import *

def init_sensor():
    """
    Initialize camera sensor with specified configuration
    使用指定配置初始化摄像头传感器
    """
    # Create sensor instance with resolution 1280x960
    # 创建分辨率为1280x960的传感器实例
    sensor = Sensor(width=1280, height=960)

    # Reset sensor to default state
    # 将传感器重置为默认状态
    sensor.reset()

    # Configure channel 1 output format to 640x480 RGB565
    # 配置通道1输出格式为640x480 RGB565
    sensor.set_framesize(width=640, height=480, chn=CAM_CHN_ID_1)
    sensor.set_pixformat(Sensor.RGB565, chn=CAM_CHN_ID_1)

    sensor.set_hmirror(True)

    return sensor
```

```python
def main():
    """
    Main function to run camera preview
    运行摄像头预览的主函数
    """
    sensor = None

    try:
        # Initialize camera sensor
        # 初始化摄像头传感器
        sensor = init_sensor()

        # Initialize virtual display with 640x480 resolution
        # 初始化640x480分辨率的虚拟显示
        Display.init(Display.VIRT, width=640, height=480, to_ide=True)



        # Initialize media management
        # 初始化媒体管理
        MediaManager.init()

        # Start sensor operation
        # 启动传感器运行
        sensor.run()

        # Main loop to capture and display frames
        # 捕获和显示帧的主循环
        while True:
            # Capture frame from channel 1
            # 从通道1捕获帧
            img = sensor.snapshot(chn=CAM_CHN_ID_1)

            # Display captured frame
            # 显示捕获的帧
            Display.show_image(img)

    except KeyboardInterrupt:
        print("User interrupted the program")
        print("用户中断了程序")

    except Exception as e:
        print(f"An error occurred: {str(e)}")
        print(f"发生错误: {str(e)}")

    finally:
        # Cleanup section
        # 清理部分

        # Stop sensor if initialized
        # 如果传感器已初始化则停止
        if isinstance(sensor, Sensor):
            sensor.stop()

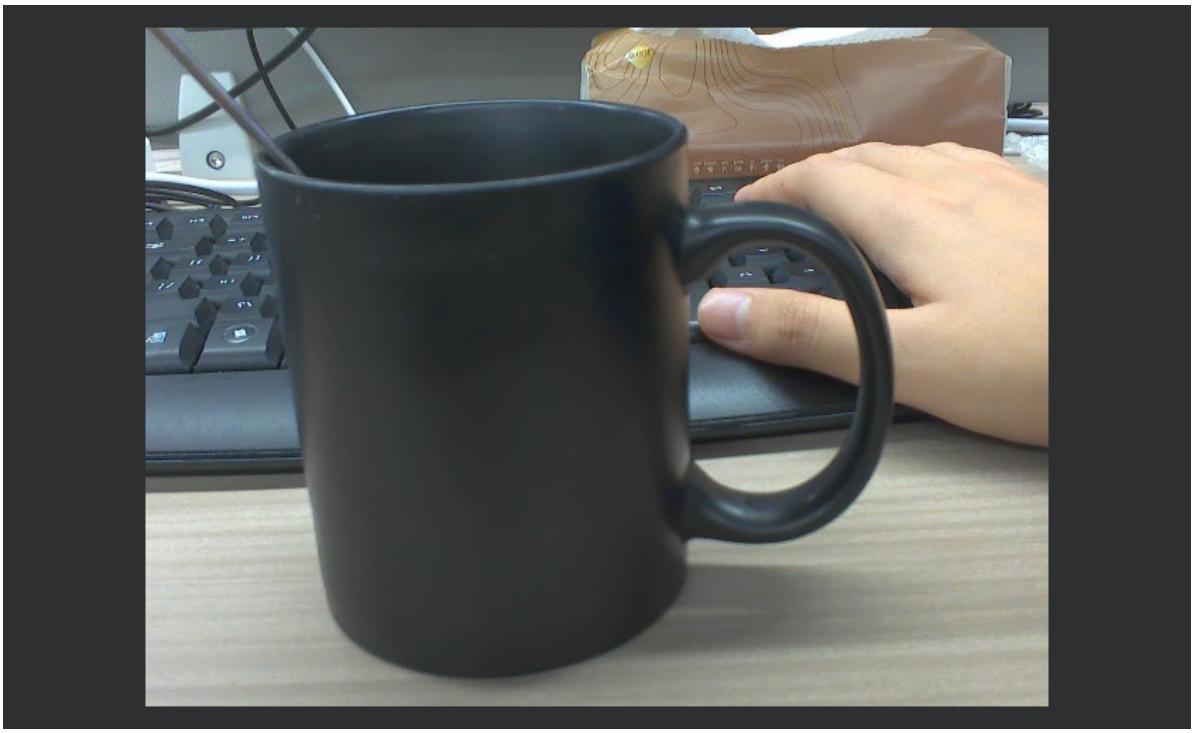
        # Deinitialize display
        # 反初始化显示
        Display.deinit()
```

```
        # Enable sleep mode
        # 启用睡眠模式
        os.exitpoint(os.EXITPOINT_ENABLE_SLEEP)
        time.sleep_ms(100)

        # Release media resources
        # 释放媒体资源
        MediaManager.deinit()

if __name__ == "__main__":
    main()
```

例程执行效果：



摄像头垂直翻转

API 描述

sensor.set_vflip

描述

配置图像传感器是否进行垂直翻转。

语法

```
sensor.set_vflip(enable)
```

参数

参数名称	描述	输入/输出
enable	`True` 开启垂直翻转功能 `False` 关闭垂直翻转功能	输入

返回值

返回值	描述
无	

举例

```
sensor.set_vflip(True)
```

例程

代码位于 [源码汇总 / 02.Basic /18.2_sensor_vflip.py]

```
"""
Camera preview demo
摄像头预览演示

This script initializes camera sensor, displays preview and handles cleanup
本脚本初始化摄像头传感器、显示预览并处理清理工作
"""

import sys
import uos as os
import time
from media.sensor import *
from media.display import *
from media.media import *

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    """
    Initialize camera sensor with specified configuration
    使用指定配置初始化摄像头传感器
    """
    # Create sensor instance with resolution 1280x960
    # 创建分辨率为1280x960的传感器实例
    sensor = Sensor(width=1280, height=960)

    # Reset sensor to default state
    # 将传感器重置为默认状态
    sensor.reset()

    # Configure channel 1 output format to 640x480 RGB565
    # 配置通道1输出格式为640x480 RGB565
    sensor.set_framesize(width=640, height=480, chn=CAM_CHN_ID_1)
    sensor.set_pixformat(Sensor.RGB565, chn=CAM_CHN_ID_1)

    sensor.set_vflip(True)

    return sensor

def main():
    """
    Main function to run camera preview
```

```
    运行摄像头预览的主函数
    """
    sensor = None

    try:
        # Initialize camera sensor
        # 初始化摄像头传感器
        sensor = init_sensor()

        # Initialize virtual display with 640x480 resolution
        # 初始化640x480分辨率的虚拟显示
        Display.init(Display.VIRT, width=640, height=480, to_ide=True)



        # Initialize media management
        # 初始化媒体管理
        MediaManager.init()

        # Start sensor operation
        # 启动传感器运行
        sensor.run()

        # Main loop to capture and display frames
        # 捕获和显示帧的主循环
        while True:
            # Capture frame from channel 1
            # 从通道1捕获帧
            img = sensor.snapshot(chn=CAM_CHN_ID_1)

            # Display captured frame
            # 显示捕获的帧
            Display.show_image(img)

    except KeyboardInterrupt:
        print("User interrupted the program")
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    except Exception as e:
        print(f"An error occurred: {str(e)}")
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        # Stop sensor if initialized
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        if isinstance(sensor, Sensor):
            sensor.stop()

        # Deinitialize display
        # 反初始化显示
        Display.deinit()

        # Enable sleep mode
        # 启用睡眠模式
        os.exitpoint(os.EXITPOINT_ENABLE_SLEEP)
```

```
        time.sleep_ms(100)

        # Release media resources
        # 释放媒体资源
        MediaManager.deinit()

if __name__ == "__main__":
    main()
```

例程执行效果如下：

